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**FAILING IS NOT AN OPTION: THE DIGITAL BREAKTHROUGH THAT REINVENTED
FISHING OPERATIONS**

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ABSTRACT

Fishing operations represent one of the most uncertain and high-risk aspects of well intervention. In the Narok Onshore Field during 2024, over 5,000 workover and intervention activities were carried out. Among these were 245 fishing jobs. A significant challenge is that nearly 90% of these were unplanned, reactive operations with highly unpredictable success rates. This unpredictability leads directly to operational delays, increased costs, and lost production, highlighting an urgent need for better planning and execution.

Several root causes contribute to this problem. Currently, there are no predictive methods to forecast fishing incidents or reliably gauge the probability of success. Decisions often depend on individual engineer experience, leading to inconsistent technical judgements. Going in depth, workflows are not standardized, critical data is scattered across different sources, and teams rely heavily on manual processes. Under this pressure, technical miscalculations and rushed economic evaluations can escalate costs and lead to operational failure.

To address these gaps, the team developed a digital solution called FOCUS (Fishing Optimization and Candidate Underpinning Strategy). This mobile-enabled system integrates historical field data with machine-learning simulations to provide adaptive, evidence-based recommendations. FOCUS creates a standardized framework that predicts potential fishing in well candidates, success probabilities for every instance of fishing job, estimates job costs and durations, suggests optimal bottom hole assembly (BHA) designs, and incorporates economic analysis.

Keywords: fishing, digitalization, workover, well intervention

INTRODUCTION

In the Narok Field, workover and well intervention activities are essential for sustaining and boosting oil production. While these operations are routine, their execution is fraught with challenges and inherent uncertainties. Operations involving running, pulling, or define downhole equipment, for example, carry a constant risk of tools becoming stuck. Resolving these situations requires specialized retrieval operations, known as fishing jobs.

The conventional approach to fishing jobs is inherently slow. When a problem occurs in the field, operations cannot start immediately. Instead, it triggers a lengthy process of coordination and meetings between operations engineers, planning teams, and service companies to develop a fishing plan. The subsequent mobilization of specialized equipment adds further delay. As noted by Harrison [1], the uncertainty inherent in fishing operations has long been recognized as a primary driver of non-productive time and operational cost overruns. This preparation gap is inefficient, especially given the scale of activity in Narok—approximately 5,000 intervention jobs in 2024, which included 245 fishing jobs with a success rate of 83%. The trend continued into early 2025, with 36 fishing jobs identified out of 1,038 interventions, achieving an 100% success rate. This shift is designed to close the preparation gap and deliver more consistent, successful outcomes in fishing operations.

METHODS

Data Collection

The development of the FOCUS system started with gathering comprehensive historical data (Figure 1). For each well identified as a potential fishing candidate, we collected all relevant job histories, well schematics, completion details, and downhole equipment records. This was an iterative process. The completeness of each dataset was rigorously checked; if gaps were found, the collection process was repeated for that well. Only when a dataset fully met our criteria did we move forward. This validated data then served as the foundation for all subsequent steps: integrated analysis, well program development, fishing job recommendations, economic evaluation, and job duration forecasting.

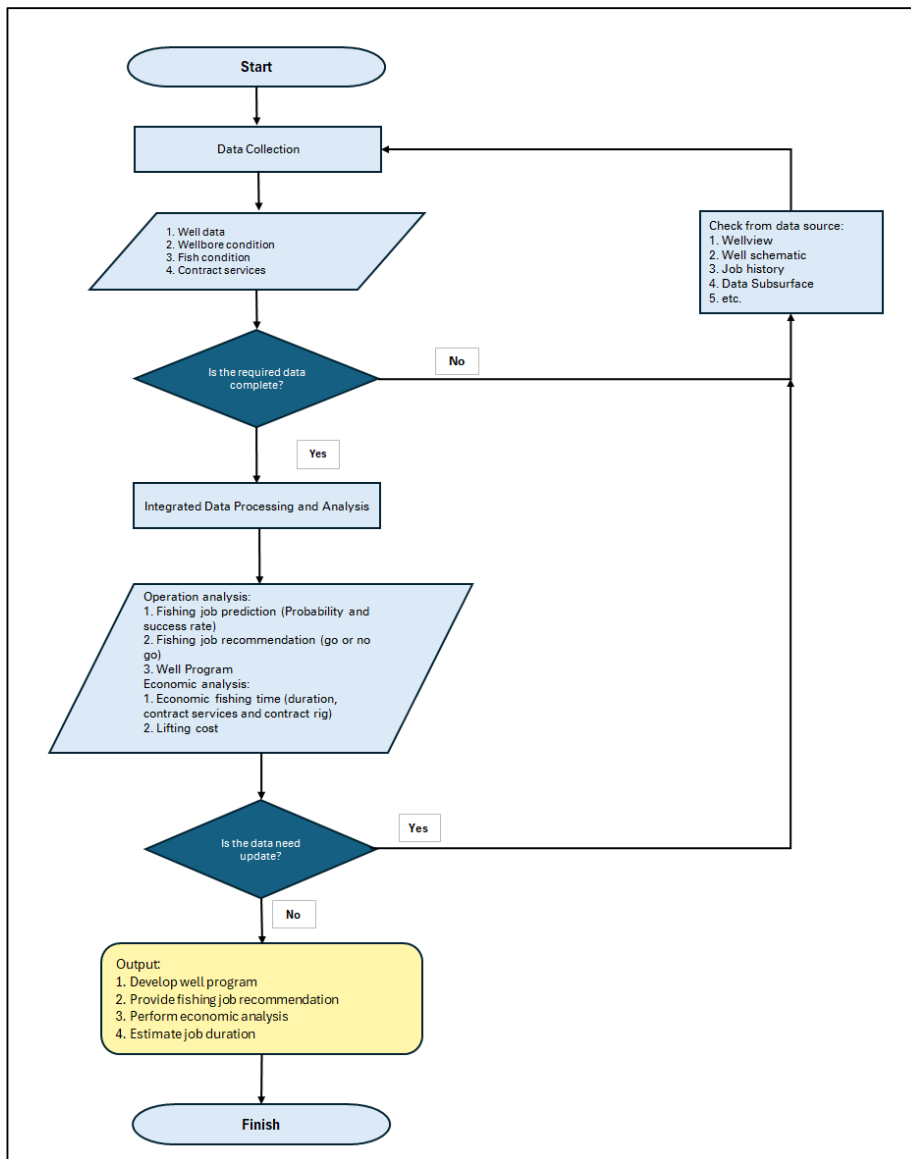


Figure 1 Flow Chart Data Collection and Processing

Methodology for Field Condition

Field condition risk modeling

Field condition represents the macro-level operational exposure that establishes the prior probability of fishing difficulty within the Fishing Hybrid Decision Algorithm (FHDA).

The Field Risk Index (R_{field}) is quantified specifically for each operational area or field by integrating actual subsurface data, historical records, statistical trends, and typical well behavior. This approach enables the formulation of several variables that equals to a unique variable complexity for each field, defined as:

$$R_{\text{field}} = V_{r1} \times V_{r2} \times \dots \times V_{rn}$$

V_{r1} = Variable Risk 1

V_{r2} = Variable Risk 2

V_{rn} = Variable Risk n

For Narok*, the dominant risk variables influencing operational complexity are fluid-loss-related issues and sand production tendencies. Consequently, the field-specific risk index is expressed as:

$$R_{\text{field}}^* = VCI^* = WFLI^* \cdot SPI^*$$

WFLI = Well Fluid Loss Index

SPI = Sand Problem Index

* Narok field

**risk variable might be different for other Field or Area (such as pressure, etc)

This index serves as the FHDA prior, representing baseline operational difficulty before well-specific or fish-specific conditions are evaluated.

Wellbore condition risk modeling

Wellbore condition refines the field-level prior by incorporating structural, mechanical, and historical well attributes that influence fishing complexity. Each wellbore parameter is encoded as a normalized risk contributor, reflecting its empirical tendency to escalate fishing operations. The wellbore risk index (R_{wellbore}) is defined as:

$$R_{\text{wellbore}} = \left(\sum_{i=1}^n \lambda_i \cdot w_i \right)^{\beta}$$

λ_i = Parameter sensitivity

β = Nonlinearity factor (set to 1 for base FHDA implementation)

w_i = Wellbore Index

The index captures combined effects of abandonment history, geometry, formation exposure, downhole tools object characteristics, casing integrity, and hole condition. This formulation enables FHDA to differentiate structurally simple wells from mechanically complex wells under identical field conditions.

Fish condition risk modeling

Fish condition represents the micro-level retrievability uncertainty, capturing the quality and completeness of information available about the downhole object.

Two dominant variables are considered is fish detail information completeness (f_1) and top of fish identification clarity (f_2). The Fish Risk Index (FRI) is defined as:

$$R_{\text{fish}} = f_1 \times f_2$$

Lower FRI values indicate increased uncertainty in engagement, alignment, and recovery, directly affecting fishing success probability.

Downhole Tools Factor modeling (DTF)

The Downhole Tools Condition (DTF) represents the computed condition state of downhole tools, derived from multiple influencing factors including tool type, material properties, duration of exposure within the wellbore, and the specific configuration in which the tool is deployed. These parameters collectively define the degradation profile and operational readiness of each downhole tool, enabling a more accurate assessment of its reliability and associated operational risk.

Fishing performance history modeling

Fishing performance history (FPH) is derived from historical fishing-operation data specific to the types of fish encountered in the Narok Field. The dataset is compiled, analyzed, and statistically processed using thousands of fishing events recorded. The resulting FPH value is expressed as a percentage, representing the historical success rate and efficiency associated with resolving each specific fish type.

This hierarchical structure ensures that field exposure, wellbore complexity, and fish uncertainty are all explicitly accounted for, enabling defensible, explainable fishing strategy selection

Operational Analysis

Potential fishing

The "Potential Fishing" or "First Entry" mode is used during the initial screening and planning phase for well interventions. In this stage, engineers assess a well to determine if the downhole equipment and current wellbore conditions on specific wells present a risk of becoming a "fish". An object that could get stuck or lost. This evaluation is mathematically grounded in the FHDA's risk synthesis model, which calculates a prior probability of a fishing event. The system integrates macro and micro-level risks using the formula,

$$\text{Potential Fishing} = R_{\text{field}} \times R_{\text{wellbore}} \times DTF$$

R_{field} = risk index

R_{wellbore} = wellbore index

DTF= downhole tools condition

The Potential Fishing percentage represents the estimated likelihood that a fishing event will be required when conducting a well intervention on a specific candidate well. This is a pre-job risk assessment calculated during the initial screening and planning phase ("First Entry" mode) to answer: "What is the probability that this specific well intervention will result in downhole equipment becoming stuck or lost, requiring a fishing operation?"

Success ratio

The "Success Ratio" or "Not First Entry" mode is activated once a fishing job has been identified and is about to commence. This focuses on the confidence level of successfully retrieving an existing fish from the wellbore. The system shifts from risk prediction to success probability estimation by leveraging the FHDA's multi-layer decision framework. It computes a composite decision score

$$Success\ Ratio = R_{field} \times R_{wellbore} \times R_{fish} \times FPH$$

R_{field} = risk index

$R_{wellbore}$ = wellbore index

R_{fish} = fishing index

FPH = fishing performance index

This score is then translated into a success probability percentage. Crucially, this feeds directly into the Economic Fishing Time (EFT). Where R_{field} represents Risk Field Index, $R_{wellbore}$ accounts for casing and geometry conditions, and the R_{fish} factors evaluate the available information of the fish and the fishing performance history (FPH).

Planned fishing cost

Planned Fishing Cost represents a calculated estimate derived from the historical cost records of fishing operations associated with specific fish types. This calculation integrates detailed cost data from past fishing events with the contractual pricing structure of the fishing services available in the Narok Field. By combining historical expenditure patterns with contract-based cost components, the planned fishing cost provides a field-specific, data-driven projection of the expected financial impact should a fishing event occur.

Planned cycle time

Planned cycle time is derived from the historical performance data of fishing operations conducted in the Narok Field. This estimate is generated by compiling and analyzing thousands of past fishing activities, incorporating detailed operational parameters as part of the computation. By integrating these granular performance factors, the planned

fishing time provides a data-driven projection of the expected duration required to resolve a specific type of fish under comparable operational conditions.

Economic analysis

Loss Production Oil

Lost Production Oil is one of the most important factors to consider when evaluating economic fishing time. Lost oil production is calculated using several variables, including the agreed-upon production sharing percentage under the Operator-Government agreement (Gross Split or Cost Recovery), the well's production rate, the well's decline rate, and the fishing costs (rig and fishing contract costs).

$$\text{Estimate Revenue (ER)} = (DP \cdot EDP \cdot DRI \cdot \text{Oil Price} \cdot \text{ShP}) - (LC \cdot EDP \cdot DP)$$

ER = Estimated revenue if the well successfully returns to production after the fishing job.

DP = Daily Production (Bbls)

EDP = Estimate days of production (Days)

DRI = Decline Rate Index (%)

ShP = Sharing agreement between Operator and Government (%)

LC = Lifting Cost (\$/bbls)

Loss oil Production can be defined by cumulative of Estimate Revenue (ER).

$$\text{Loss Production Oil (TLPO)} = \sum_{i=1}^n ER$$

LPO = Loss Production Oil (\$)

ER = estimated revenue if the well successfully returns to production after the fishing job.

Economic Fishing Time

The wellbore condition and fish condition each have a contribution scale to the success of the fishing operation, represented in percentage (%). Duration calculations are also performed to evaluate the effectiveness of the operation when executed, using the following equation to estimate potential fishing and success ratio:

$$\text{Economic Fishing Time (EFT)} = \frac{Ps \times LPO}{Pjc + DFC}$$

EFT = economic fishing time (days)

FPs = Fishing Potential (%)

LPO = lost production oil (\$)

Pjc = previous job cost (\$)

DFC = daily fishing cost (\$/day)

This economic modeling approach builds upon foundational work by Keller et al. [2], who first established the relationship between time limitations and fishing economics, and more recent applications of artificial intelligence by Khajehniyazi [3] for optimizing fishing time decisions.

Integrated Data Processing and Analysis

The F.O.C.U.S Hybrid Decision Architecture (FHDA) is an interpretable, multi-layer decision-support system for fishing operations that integrates expert judgement, historical data, and risk-based evaluation. It converts operational inputs into normalised indicators of feasibility, complexity, and execution risk. Instead of fixed pass/fail thresholds, FHDA treats decisions as a graded optimisation problem under uncertainty, enabling evaluation of marginal and higher-risk cases. In its first stage, FHDA applies fuzzy inference to assess individual operational variables. such as fish type, restriction severity, and well geometry, assigning each a feasibility score between 0 and 1 that reflects partial compliance with best-practice criteria.

The aggregated technical feasibility score is computed as a weighted combination of individual memberships, conceptually represented as

$$F = \sum_{i=1}^n w_i \cdot \mu_i$$

Where μ_i represents the fuzzy membership of parameter i , and w_i represents its expert-defined relevance. This approach enables the algorithm to capture expert intuition while remaining mathematically transparent. The program analyses the current instance to a database of previous fishing occurrences, assigning similarity ratings based on shared characteristics such fish geometry, depth, and operational sequence.

H (Historical Success Probability): A value representing the likelihood of success based on similar cases in the database.

$$H = \frac{\sum_{j=1}^m s_j \cdot r_j}{\sum_{j=1}^m s_j}$$

where s_j is the similarity score of historical case j and r_j represents its success indicator. In parallel, FHDA performs multi-criteria decision analysis to integrate technical

feasibility, historical confidence, and operational risk into a single decision metric. Risk is treated explicitly as a modifying factor rather than a post-processing filter.

The decision score can be conceptually represented as

$$D = F \cdot H \cdot (1 - R)$$

This is the final "Command Center" formula that determines the Go or No Go recommendation.

- D (Decision Score): The final output used to rank the viability of the operation.
- R (Normalized Risk Index): A penalty factor (0 to 1). A high R (approaching 1) represents unacceptable safety or complexity risks, which will mathematically "zero out" the decision score regardless of how high F or H are.
- $(1 - R)$ (Safety Multiplier): This represents the "safe operating margin". If risk is 20% (0.2), the multiplier is 0.8, slightly reducing the overall confidence score to account for uncertainty.

Finally, FHDA applies risk-aware decisions to generate actionable recommendations. Decision outcomes are classified into approval, conditional approval, or no-go categories based on predefined confidence and risk thresholds. Importantly, the algorithm outputs not only the recommendation but also the contributing factors, enabling engineers to understand the rationale behind each decision.

RESULTS

The FOCUS application implements a deterministic decision tree to generate a final operational recommendation of Go or No Go, taking into account economic viability, historical success likelihood, and preemptive risk screening. The choice is made based on a series of tests against predetermined operational and economic baselines, rather than a single composite score. The main regulating equation is the Economic Fishing Time (EFT) formula.

Pre-emptive Risk Gate (Potential Fishing Percentage)

The third gate employs the Potential Fishing Percentage from the "First Entry" screening mode to identify exceptionally high-risk scenarios before they progress to a full fishing task.

- A baseline threshold is established; if the Potential Fishing Percentage exceeds it, the intervention is deemed unsuitable and a modified or alternative approach is recommended.
- Two catastrophic wellbore situations (casing separated or casing damage) cause the Potential Fishing Percentage to reach 100%, resulting in an instant No-Go recommendation.

When these circumstances are met, the algorithm recognizes that a traditional fishing operation is almost certain to be required and extremely likely to fail. The suggestion

changes to require a full well integrity repair or a sidetrack operation prior to any retrieval attempt.

Technical Feasibility Gate (Success Ratio vs. Baseline)

The second gate compares the technical prognosis to historical field performance. The calculated success ratio from the FHDA model is compared to the field's baseline success percentage. This baseline is based on historical performance (for example, the previously mentioned 83%). If is equal to or larger than the baseline, the job will pass through this gate. If it is less than the baseline, the suggestion is "No Go." This means that the current operation's anticipated success is lower than the field average, indicating a higher-than-acceptable technical risk.

Economic Viability Gate (EFT vs. Planned Cycle Time)

The first and most critical decision gate compares the Economic Fishing Time (EFT) to the Planned Cycle Time (PCT) estimated for the operation.

Table 1 Final output

Final Output	
"Go"	"No Go"
Operation aspect	
- If fishing potential < threshold, and the recommendation for fishing job is "Go" - If success ratio > threshold, then the recommendation for fishing job is "Go"	- If fishing potential > threshold, and the recommendation for fishing job is "No Go" - If success ratio < threshold, then the recommendation for fishing job is "No Go"
Economic aspect	
If $EFT \geq PCT$, the recommendation is Go, this indicates that the potential value of recovered oil ($P_s \times LPO$) justifies the estimated cost ($DFC \times PCT$).	If $EFT < PCT$, the recommendation is No-Go. The operation is predicted to cost more than the value it can recover, making it economically non-viable. This gate ensures that no fishing job is approved without a positive economic return, directly tackling the problem of cost escalation highlighted in the abstract.

BHA (Bottom Hole Assembly) Recommendation Engine

Along with the Go or No-Go decision, the algorithm generates a proposed tool string or Bottom Hole Assembly (BHA). This is powered by the FHDA's case-based reasoning layer. The system compares the present scenario's properties (fish kind, depth, and wellbore geometry) to its database of previous fishing jobs. The BHA advice is mathematically calculated from the historical example j with the highest similarity score

s_j , and the success indicator r_j is positive. The algorithm basically states: "For a scenario most similar to historical Case j , the tools that succeeded were X, Y, and Z." This gives engineers a data-driven starting point for BHA design, standardising a process that was previously subjective and experience-dependent. The case-based reasoning approach aligns with findings by Mohanna et al. [4], who demonstrated that standardized tool selection significantly improves fishing efficiency and success rates.

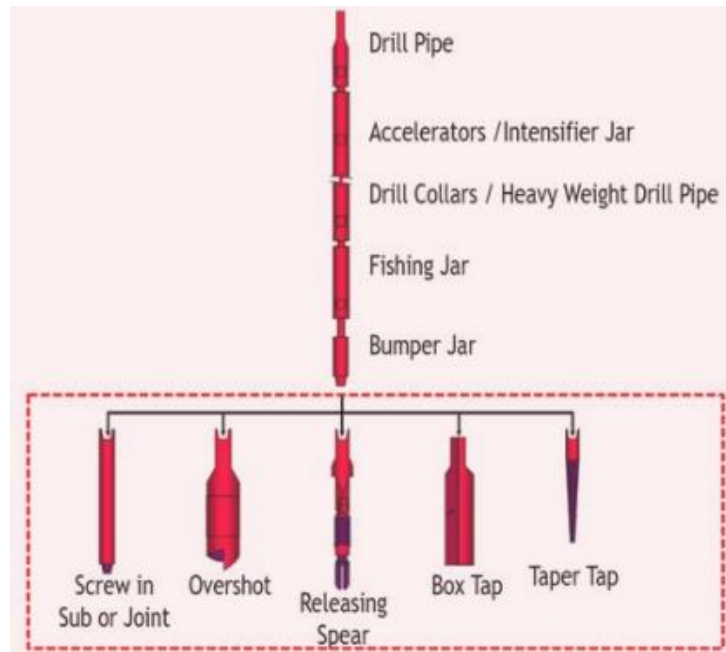


Figure 2 BHA types for fishing job

Go cases analysis

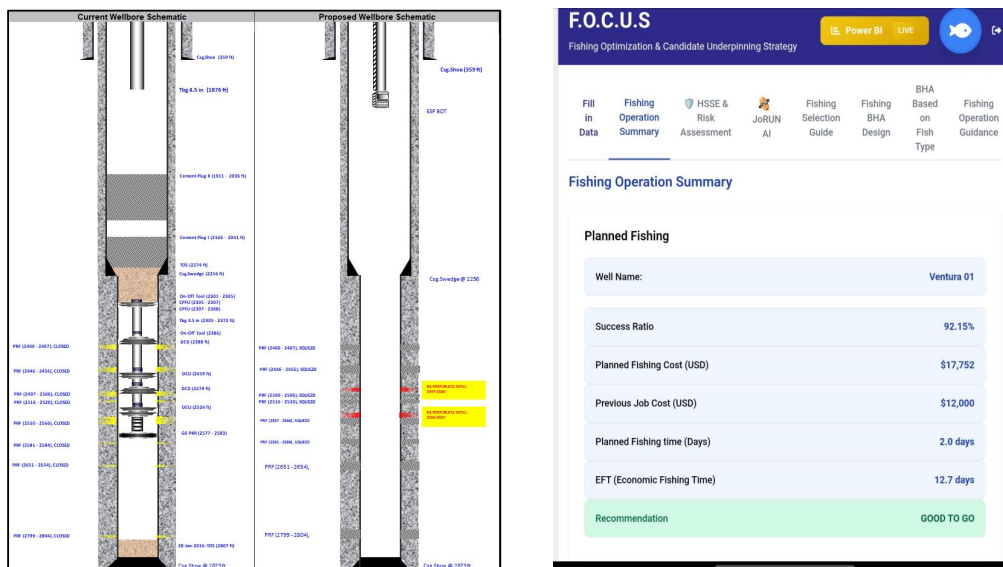


Figure 3 Well Schematic for "Go" recommendation case

Based on the evaluation conducted for Well Ventura 01, the fishing job recommendation was generated by selecting relevant field and wellbore conditions within the F.O.C.U.S system. From the field perspective, the well is located in Lando Field, which is categorized under moderate field complexity. This field condition provides a relatively predictable operational environment, allowing historical fishing performance and field behavior to be used effectively as a reference in the decision-making process.

From the wellbore condition assessment, several favorable parameters were selected. The well geometry is less than 30 degrees, indicating a low-deviation well that reduces mechanical risk during fishing operations. There is no indication of casing parted or casing damage, which significantly improves the probability of successful engagement and retrieval. Additionally, the downhole object is confirmed to have a tail length greater than 100 feet, increasing the effectiveness of conventional fishing tools. The downhole object has also been in the well for less than two years, reducing the likelihood of severe corrosion, scale buildup, or bonding effects.

By combining these field and wellbore conditions, the system was able to generate a fishing job recommendation with a high success ratio of 92.15%, an estimated fishing duration of 2.0 days, and a planned fishing cost of USD 17,752. The calculated Economic Fishing Time (EFT) of 12.7 days further supports the viability of the operation, resulting in an overall recommendation status of "GOOD TO GO."

No Go case analysis

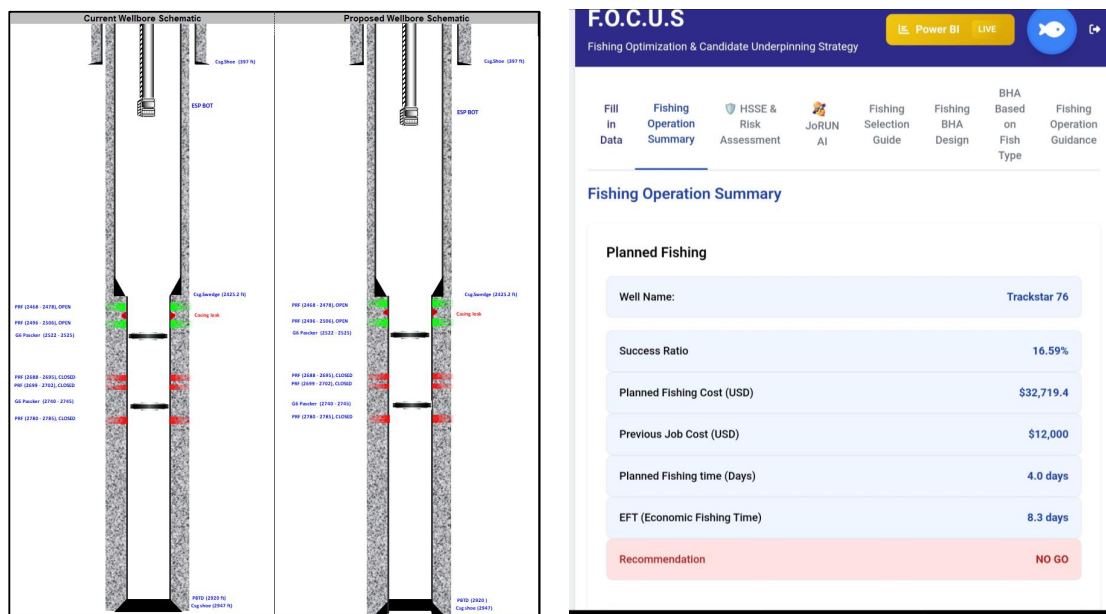


Figure 4 Well Schematic for "No Go" recommendation case

For Well Trackstar 76, the fishing job evaluation was performed by selecting both field and wellbore conditions that represent a more challenging operational scenario. From the field perspective, the well is in Narok Field, which is classified as having high field complexity. This level of complexity reflects a more uncertain operational environment,

where historical fishing outcomes indicate higher risk due to geological variability, operational constraints, and reduced predictability compared to lower-complexity fields. From the wellbore condition assessment, the selected parameters collectively contribute to an unfavorable fishing outlook. The evaluated conditions indicate increased operational risk, which significantly impacts the probability of success. As a result, the system calculated a low success ratio of 16.59%, with a planned fishing duration of 4.0 days and an estimated fishing cost of USD 32,719.4, which is substantially higher than the previous job cost of USD 12,000. Although the Economic Fishing Time (EFT) is estimated at 8.3 days, the low probability of success combined with elevated cost and extended duration leads to an overall recommendation of “NO GO.”

This outcome highlights how high field complexity, when combined with less favorable wellbore conditions, can significantly reduce fishing viability. The result reinforces the importance of risk-based decision-making, where fishing operations should be deferred or alternative mitigation strategies considered to avoid disproportionate cost exposure and low recovery likelihood. The analysis of 224 documented fishing operations based on latest data from 2025 demonstrates (Figure 5) that FHDA achieves 100% accuracy in identifying successful job interventions. While the recommendation accuracy is absolute, predictive performance for project economics shows a slight variance, with Cost Estimates and Cycle Time Estimates yielding accuracies of 76% and 73% respectively, reflecting the inherent volatility of downhole variables.

The improvement in predictability becomes particularly striking when compared to pre-FOCUS performance. Prior to system implementation, planning relied on individual engineering judgment with no standardized methodology, and anecdotal evidence suggests cost and time estimate accuracies frequently fell below 50%, with many operations experiencing cost overruns exceeding 100% of initial estimates. The step-change from sub-50% to 76%/73% accuracy represents a 25-30 percentage point improvement in estimation reliability (Figure 6), enabling far more accurate budgeting and economic decision-making before committing to operations.

The number of fishing jobs that could be executed increased significantly, reaching up to twice the actual number of fishing jobs previously completed. With an average of 18 fishing jobs per month, the implementation of the F.O.C.U.S. simulation enabled the operation to absorb 19% more jobs compared to the monthly realization, due to a shorter preparation period. As a result of this achievement, the total cost savings from fishing operations reached IDR 243 billion. Moreover, this breakthrough also contributed to a reduction in CO₂ emissions by up to 937,132.75 tons, generated from fuel consumption during fishing operations conducted by the workover rig.

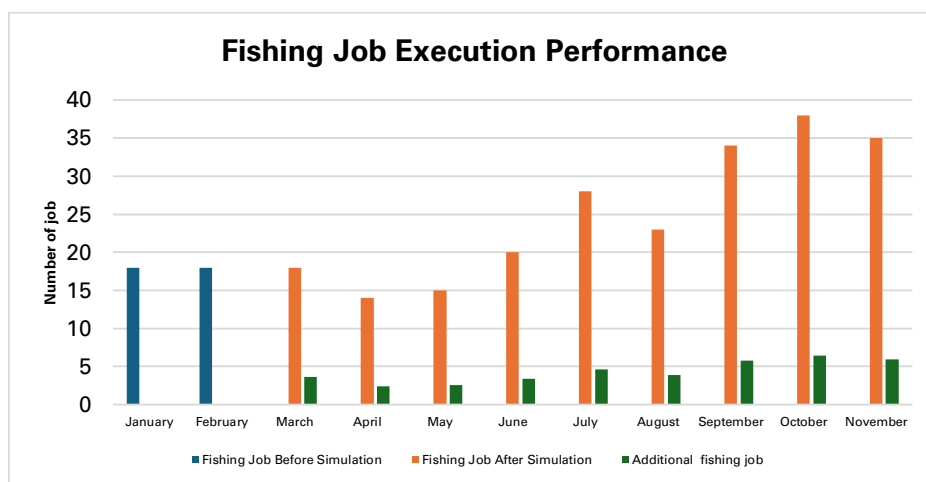


Figure 5 Fishing Job Execution Performance

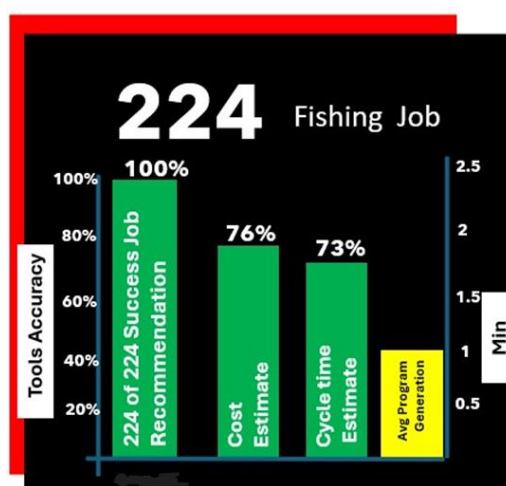


Figure 6 Economic Performance after simulation with F.O.C.U.S

The "Additional fishing jobs" (Figure 5) represent the potential upside from this workflow optimization. With program generation time reduced from hours or days to just one minute, engineers can evaluate more candidates in less time, enabling the 19% increase in monthly job throughput referenced earlier. The zeros in the "Additional" column reflect the pre-FOCUS baseline period before these efficiency gains were realized.

CONCLUSION

The implementation of the FOCUS (Fishing Optimization & Candidate Underpinning Strategy) digital platform has successfully redefined fishing operations in the Narok Onshore Field by shifting from reactive, high-risk activities to a predictable, value-generating framework. Based on 2025 performance data, the system's Fishing Hybrid Decision Algorithm (FHDA) achieved an absolute 100% accuracy in identifying

successful job interventions across 224 documented cases. This technological breakthrough facilitated an increase in the field's overall fishing success ratio from 83% to 100% while simultaneously increasing operational throughput by nearly one-third. This transformation addresses the fundamental economic principles outlined by Adkins [5], where the value of fishing operations must be evaluated against the cost of alternatives and the probability of successful recovery.

While technical recommendation accuracy is perfect, the system demonstrates realistic variance in economic forecasting, with Cost Estimate accuracy at 76% and Cycle Time Estimate accuracy at 73% due to downhole volatility. Despite these predictive variables, the platform drastically optimizes engineering efficiency by reducing the Average Program Generation time to approximately one minute, a significant improvement over traditional manual planning. These results underscore the efficacy of using machine-learning-based simulations to standardize workflows and minimize non-productive time (NPT) in complex well intervention environments.

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REFERENCE

1. Harrison, C. G., Fishing Decisions Under Uncertainty, *Journal of Petroleum Technology*, Vol. 34, No. 02, p. 299–300, 1982.
2. Keller, P., Brinkmann, P., and Taneja, P., Economic and statistical analysis of time limitations for spotting fluids and fishing operations, SPE Paper No. 4792 MS, Offshore Technology Conference, USA, Texas, Houston, 07 May 1984.
3. Khajehniyazi, S. Estimation of Optimum Fishing Time During Drilling Operation Using Artificial Intelligent Network: Gachsaran Oil Field. MS Thesis, Petroleum University of Technology, Iran, Ahwaz 2022.
4. Mohanna, A., Voghell, M., Alali, A., Bakir, R., and Al-Mousa, A., A step change in fishing efficiency: Recovering stuck pipe using the Fishing Agitation Tool, SPE Paper No. 160880 MS, SPE Saudi Arabia Section Technical Symposium and Exhibition Conference, Saudi Arabia, Al-Khobar, 08 April 2012.
5. Adkins, C., Economics of Fishing, *Journal of Petroleum Technology*, Vol. 45, No.